



CHAPTER # 6

REGISTERS AND COUNTERS

Lecturer: Noman Al Hassan
Email: noman44m@hotmail.com

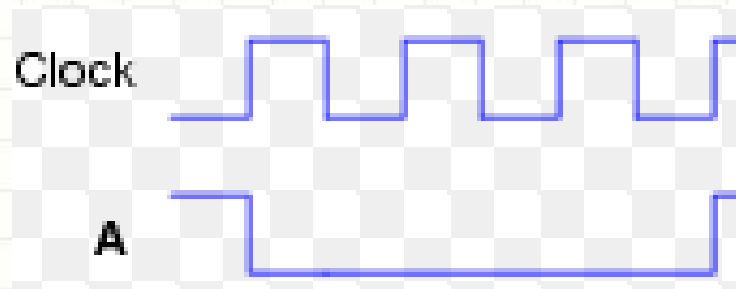
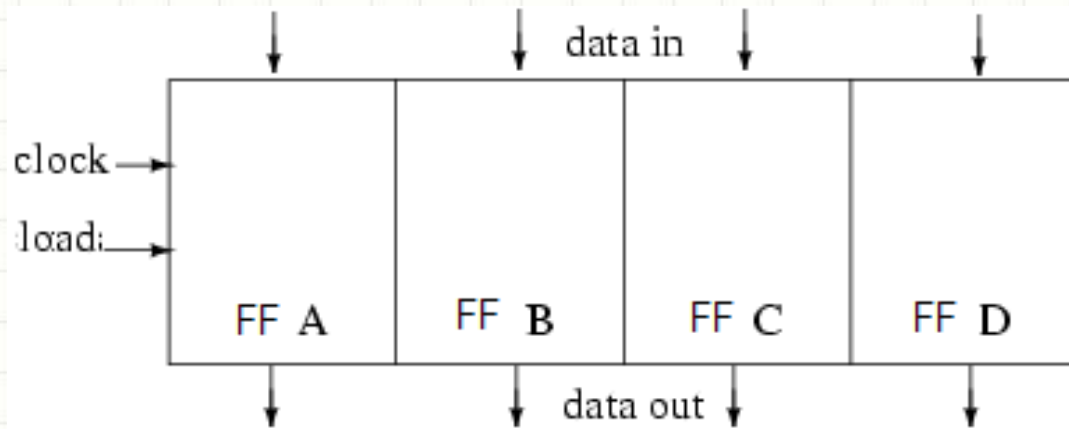
Objective of Lecture

After studying this chapter, you should be able to know

- Introduction to Registers.
- Classification of Registers.
- Shift Registers.
 - Serial Transfer
 - Serial Addition.
 - Shift right, shift left, uni directional , bi directional.
- Universal Shift Register.
- Introduction to Counters.
- Ripple counters/Asynchronous counters.
- Synchronous counters.

Introduction to REGISTERS

- To increase the storage capacity we use group of flip-flop. This group of flip-flop is known as registers.
- The n-bit register is consist of “n” numbers of flip-flop and is capable of storing “n-bit” of information.



Classification of Registers

- Registers can be classified into two forms

1. Depending upon I/P and O/P.

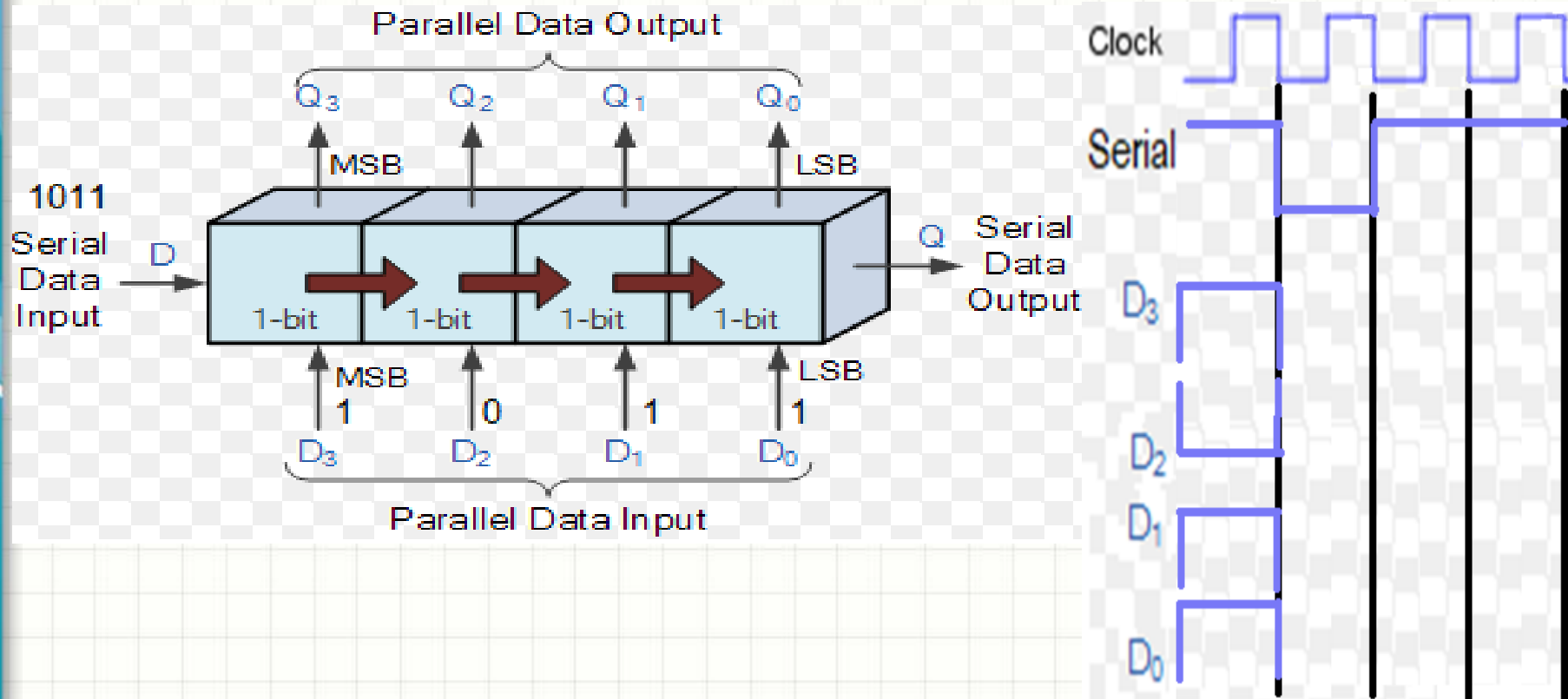
- a) SISO
- b) SIPO
- c) PISO
- d) PIPO

2. Depending upon applications.

- a) Shift Registers.
- b) Storage Registers.

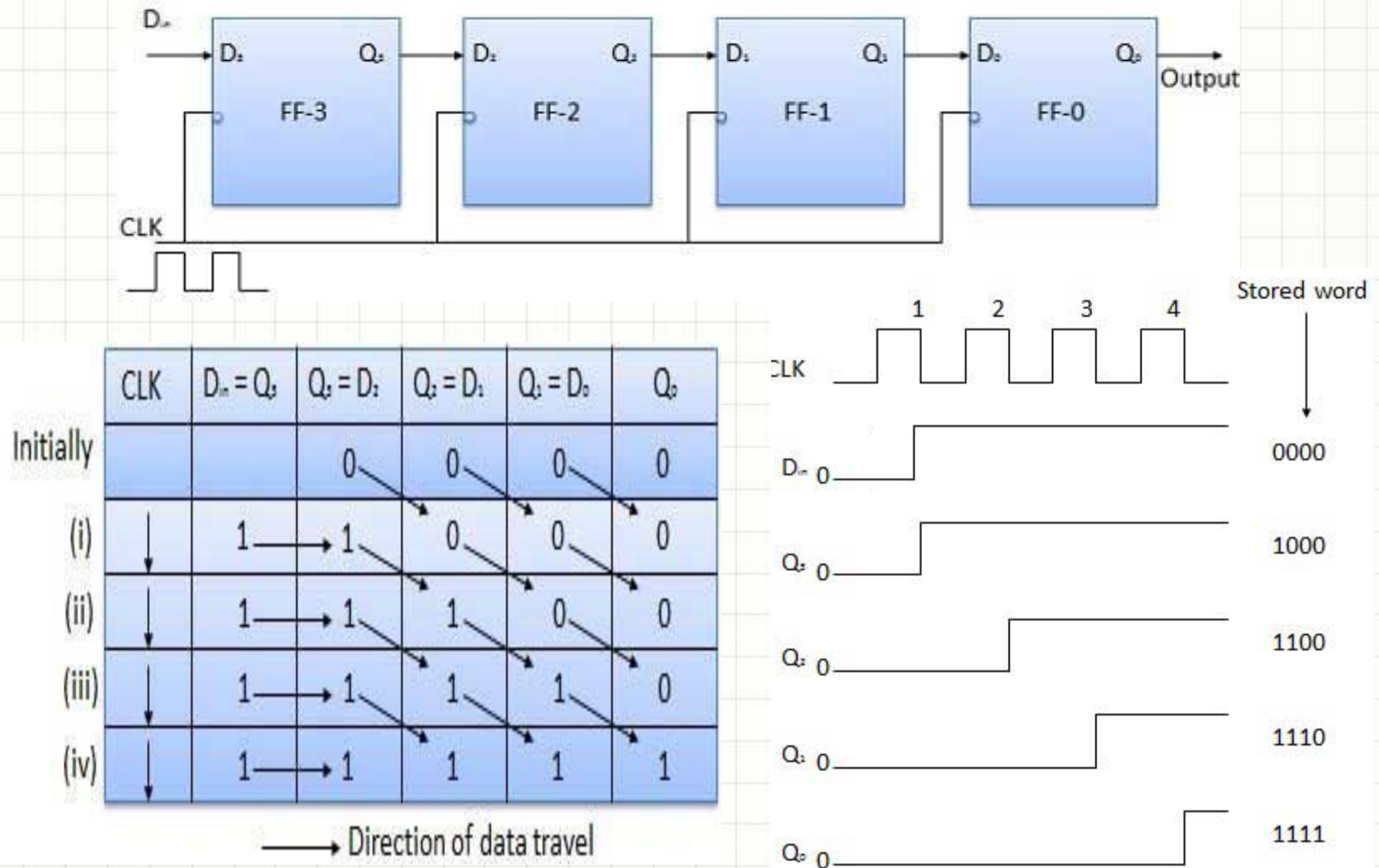
SISO and PIPO

- Data can be enter in serial or in parallel form.



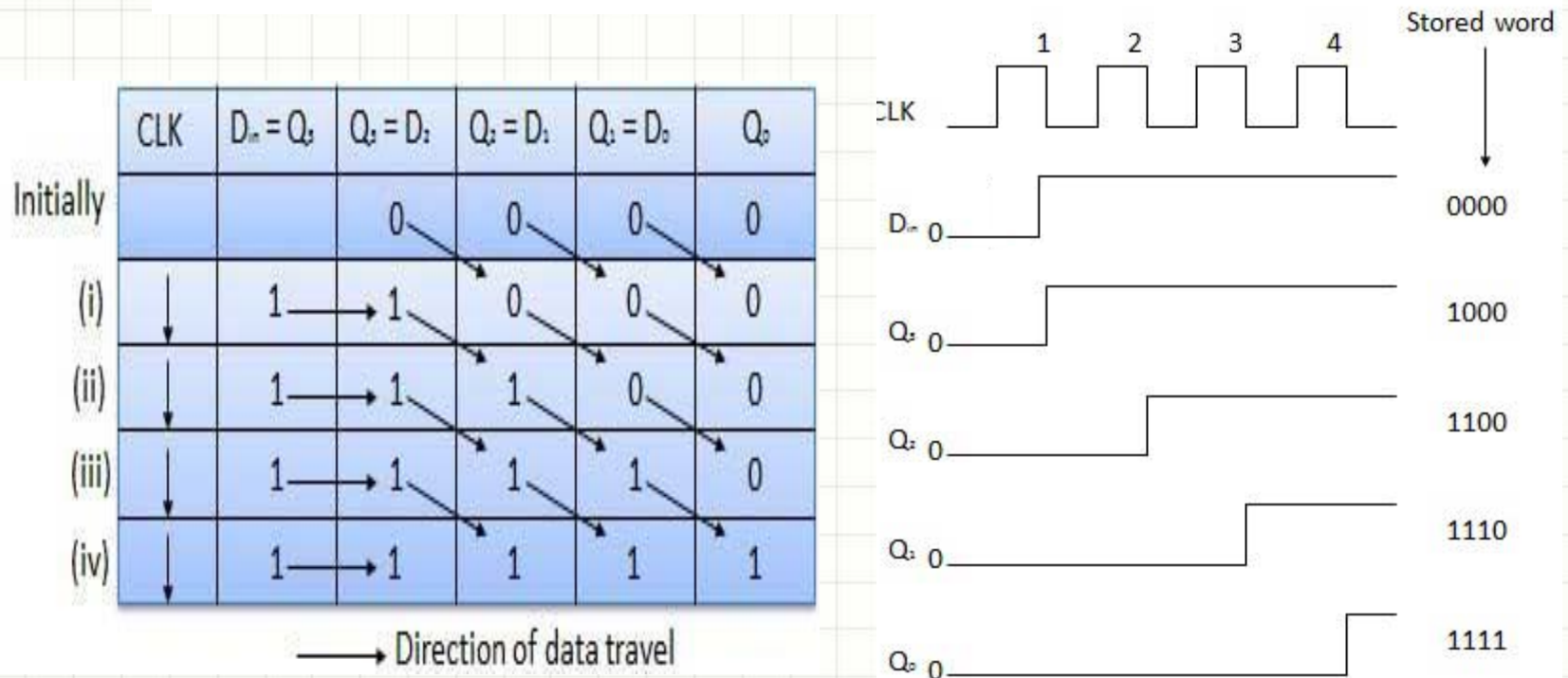
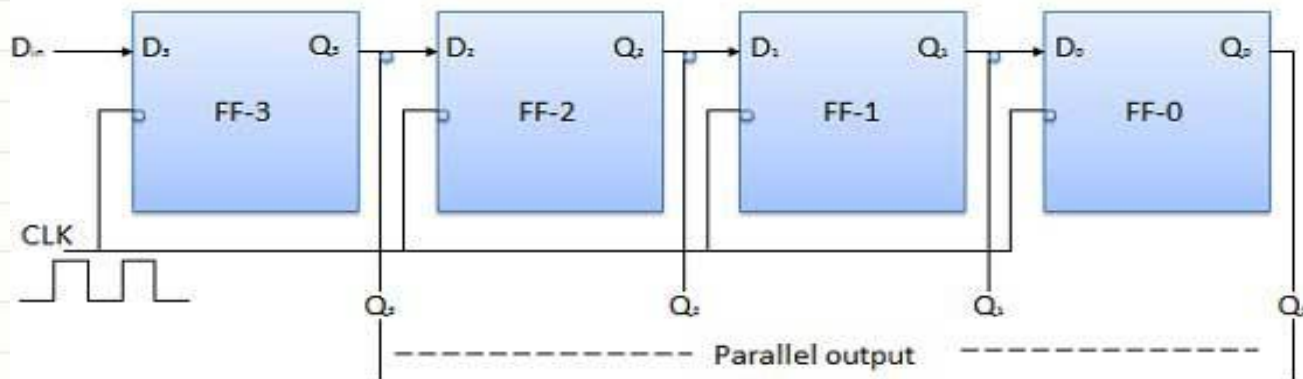
Shift Registers

- SISO



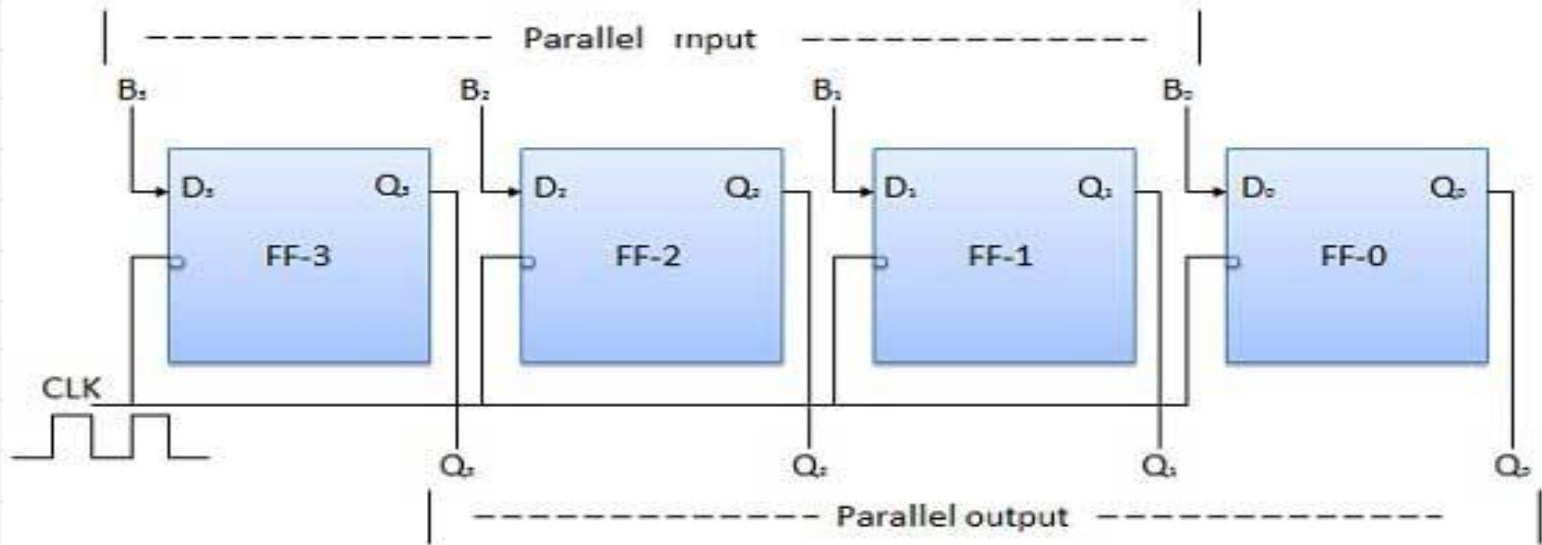
Shift Registers

- SIPO



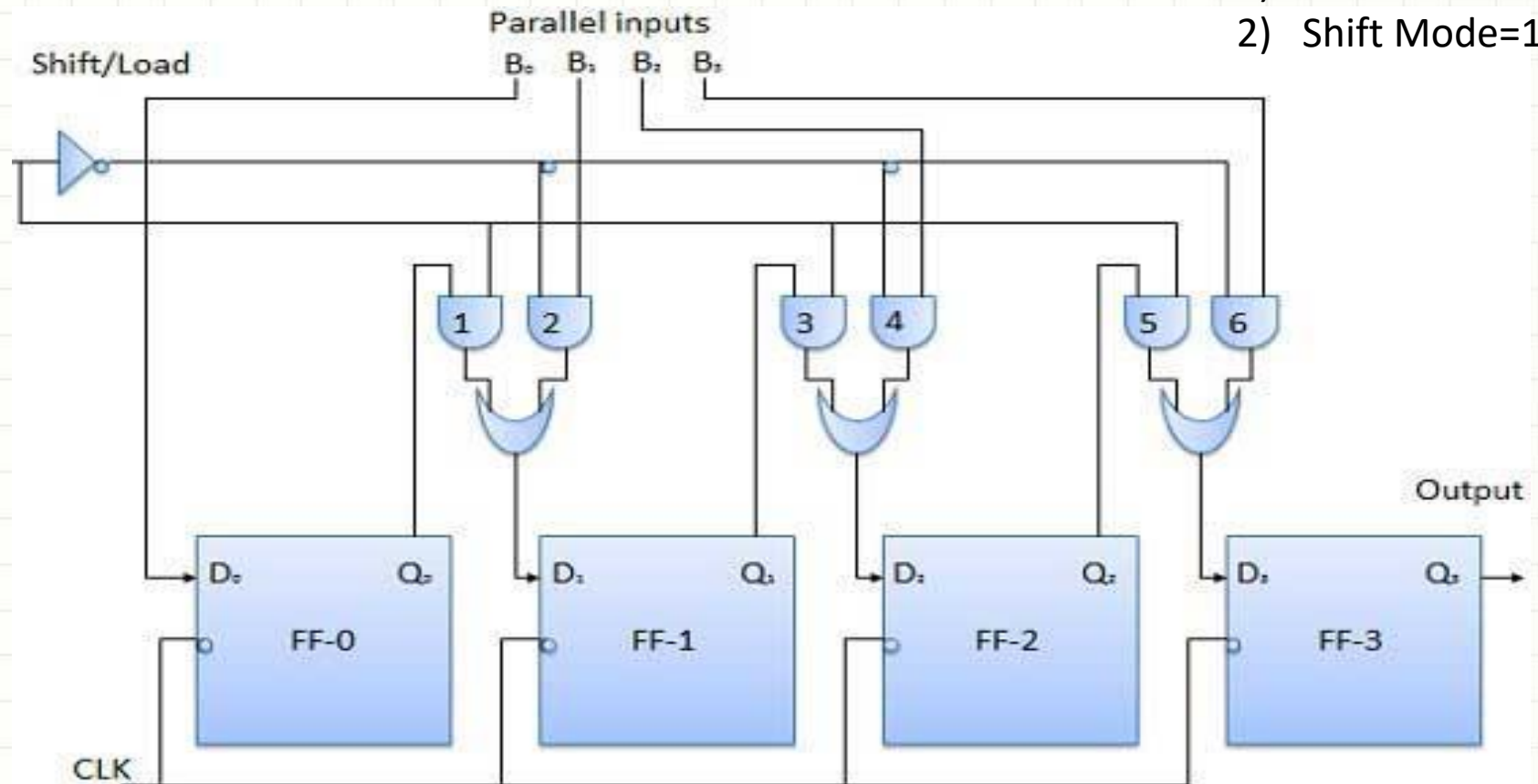
Storage Registers

- Storage register is nothing but PIPO register or buffer.



Shift Registers

- PISO

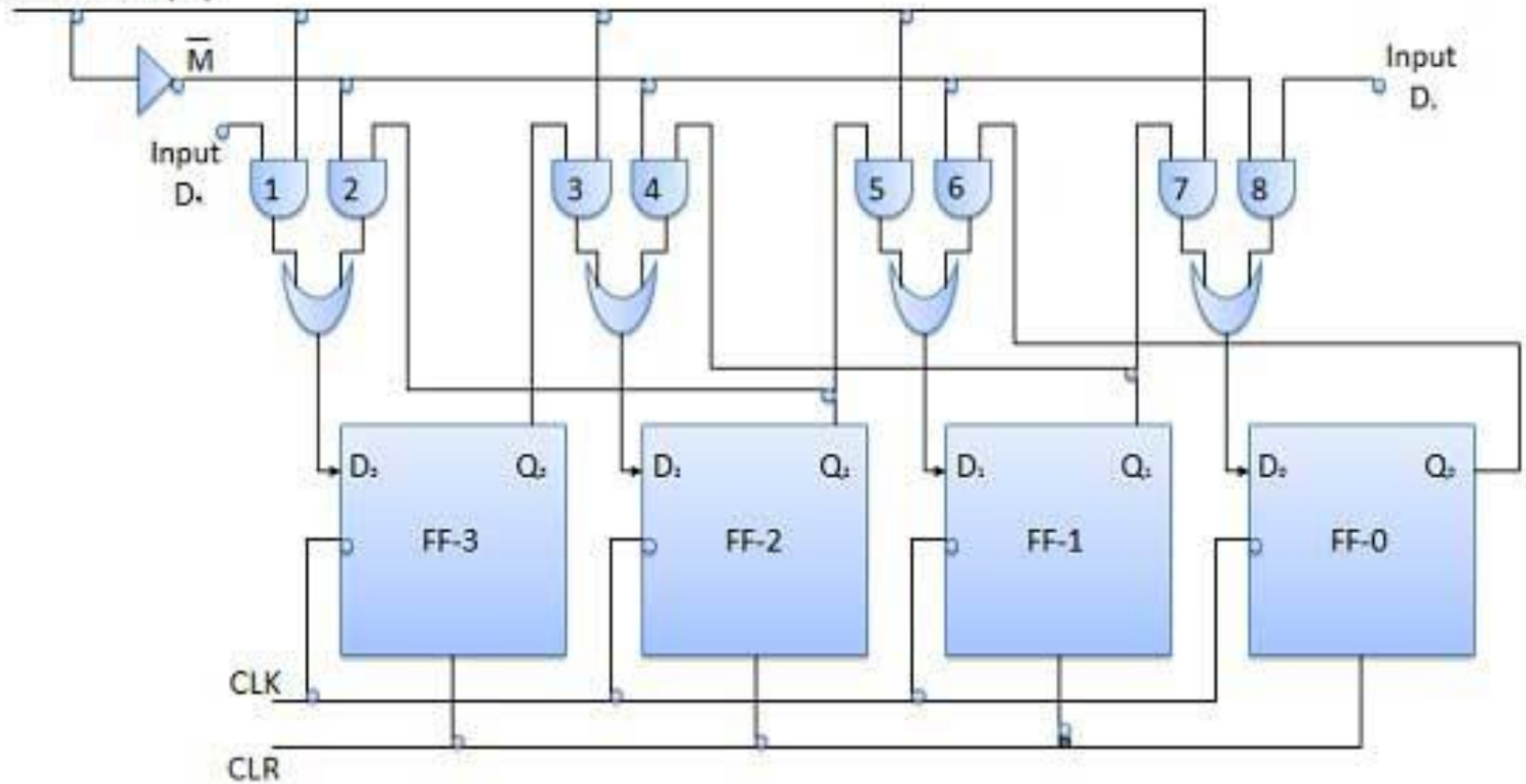


Bidirectional Shift Registers

Shift Right $M=1$

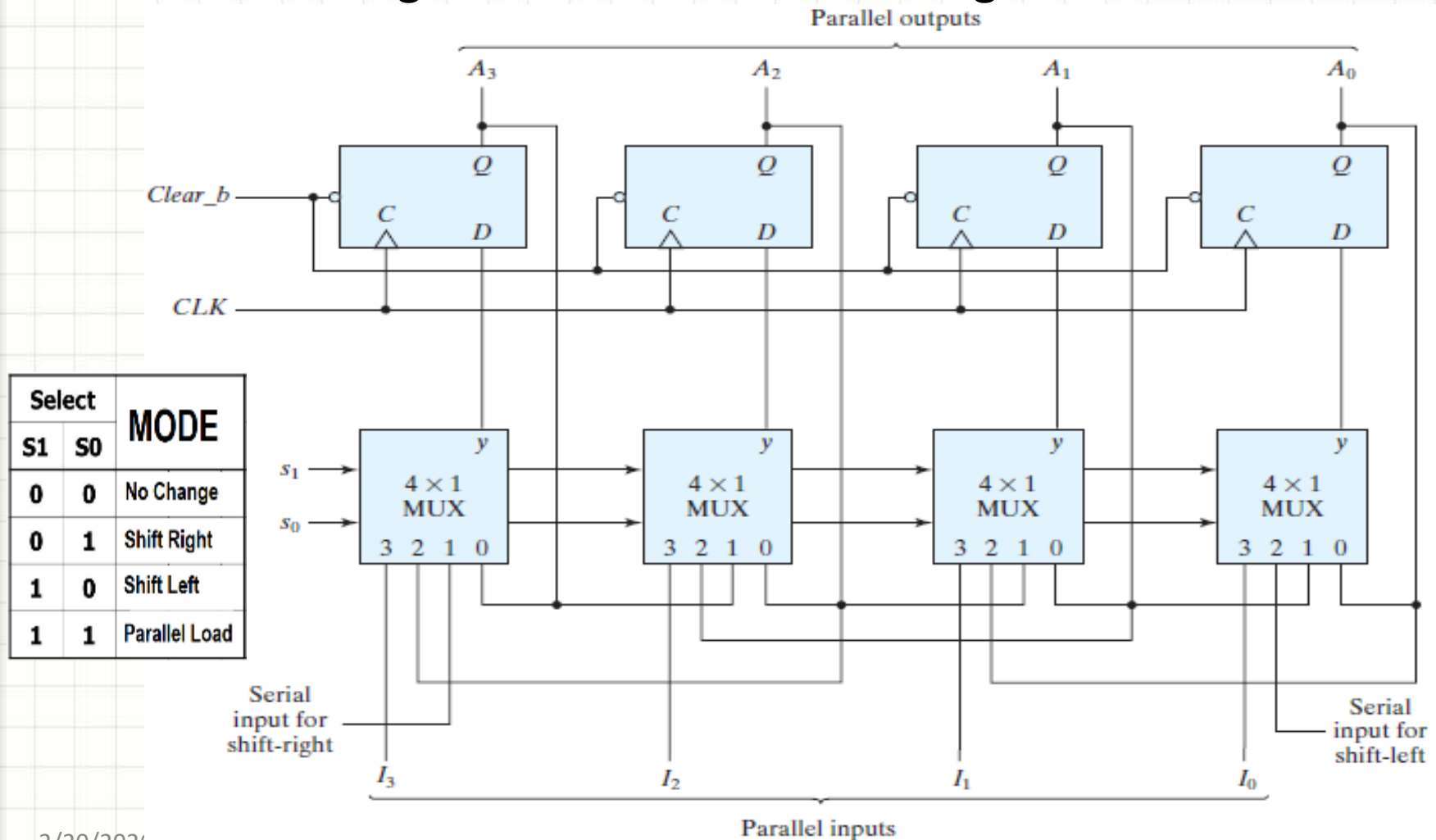
Shift Left $M=0$

Mode control (M)



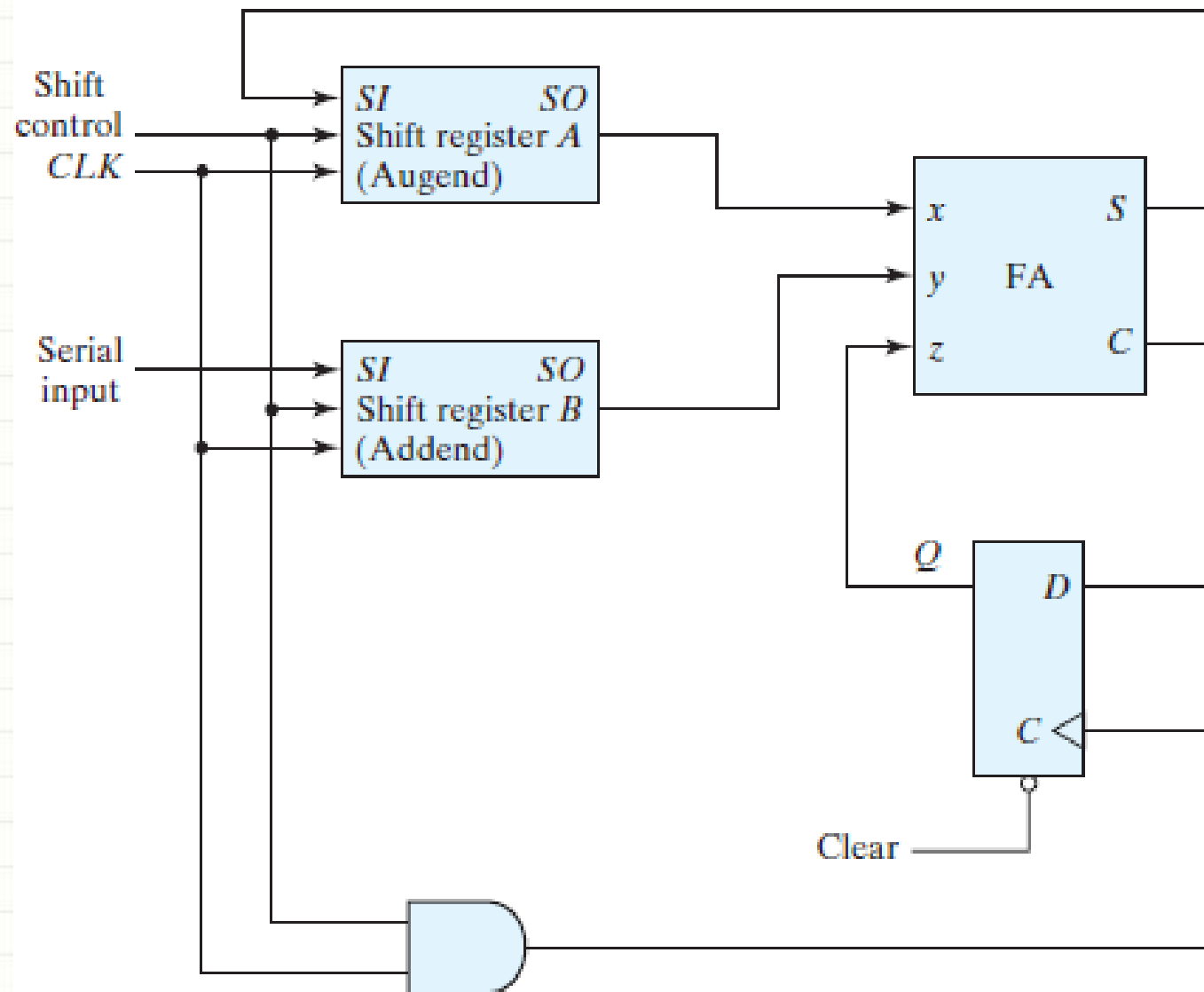
Universal Shift Register

- Unidirectional shift register or Bidirectional shift register with Parallel loading is called Universal shift register.



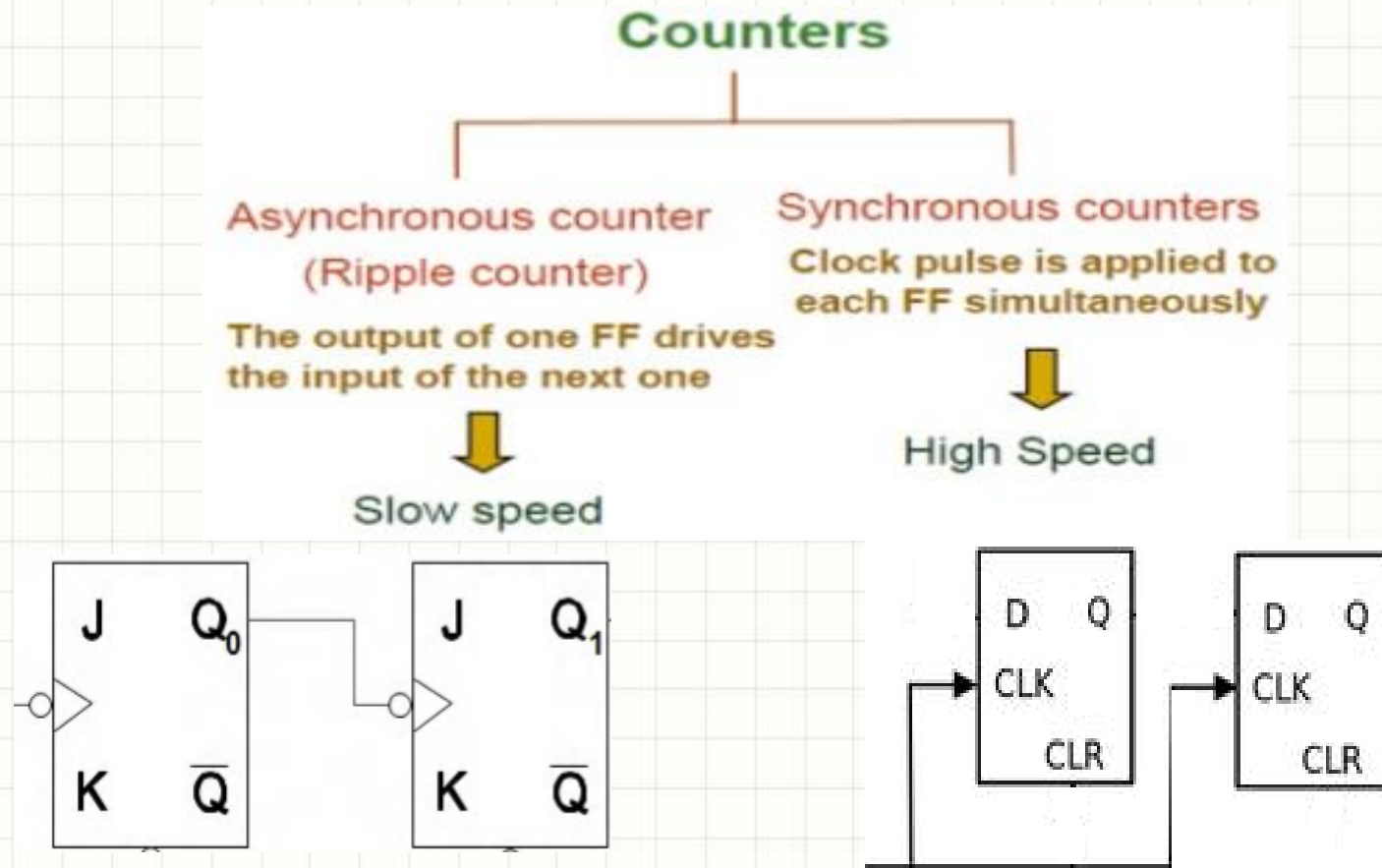
Four-bit universal shift register

Serial Addition



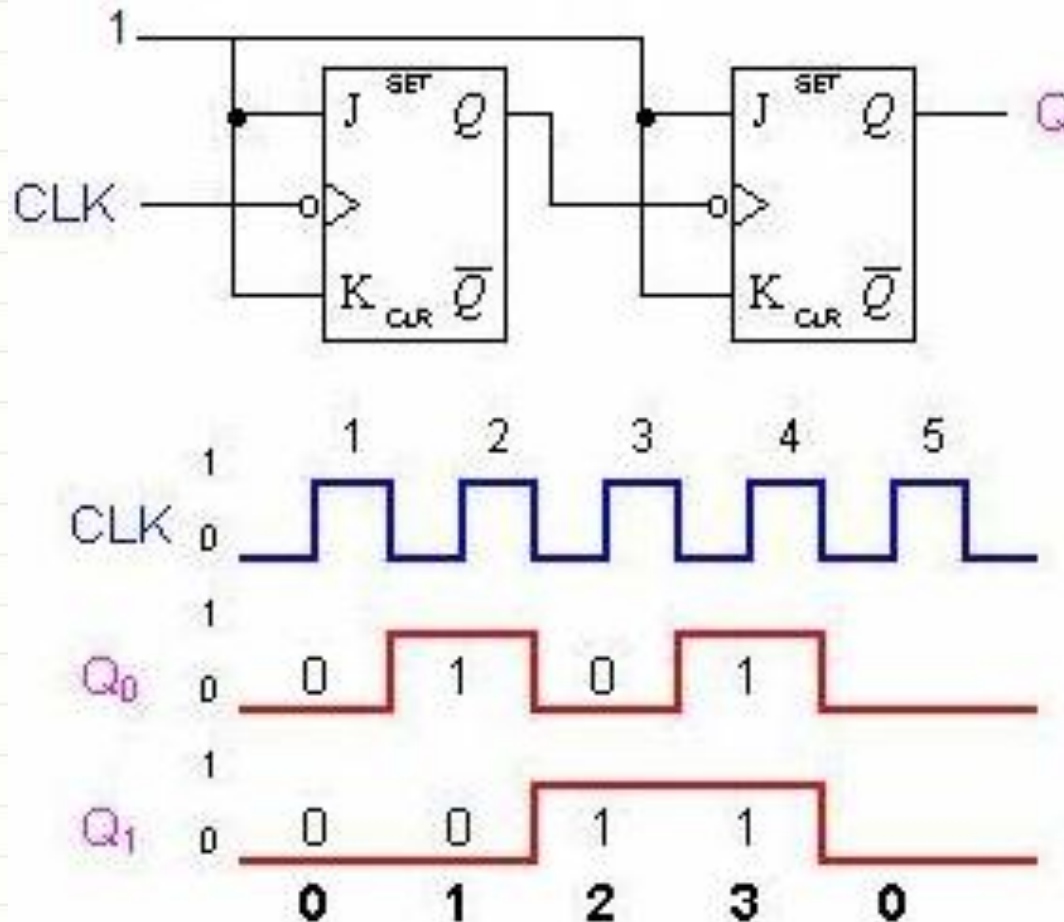
Introduction to Counters

- **Counter** is a device which stores the number of times a particular event or process has occurred. It is a sequential logic circuit consist of flip-flops



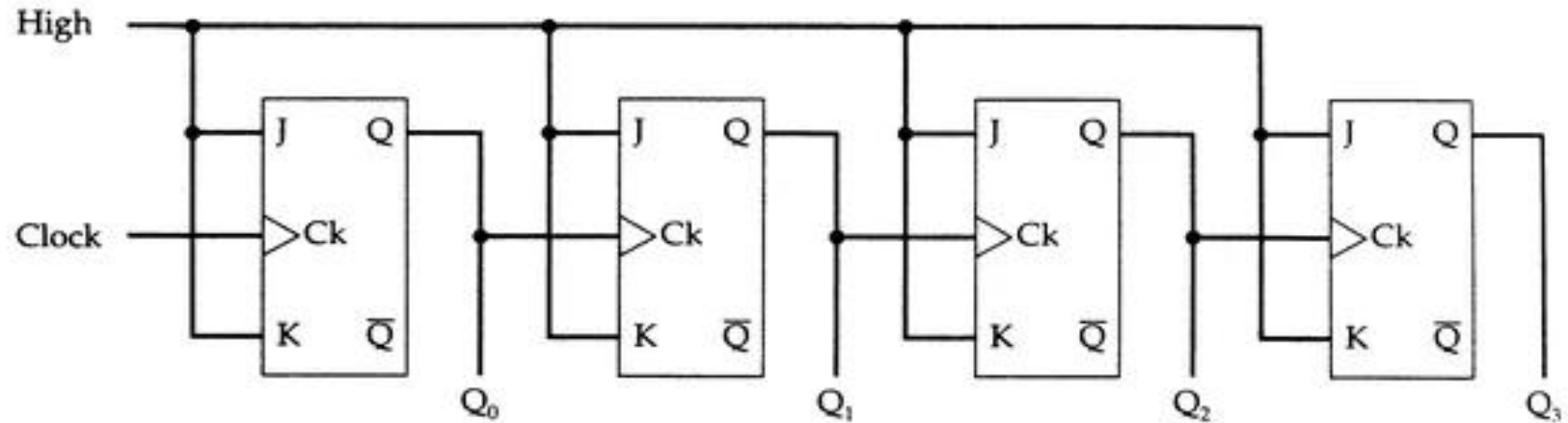
Ripple counters/Asynchronous counters

- 2 bit asynchronous up counter.

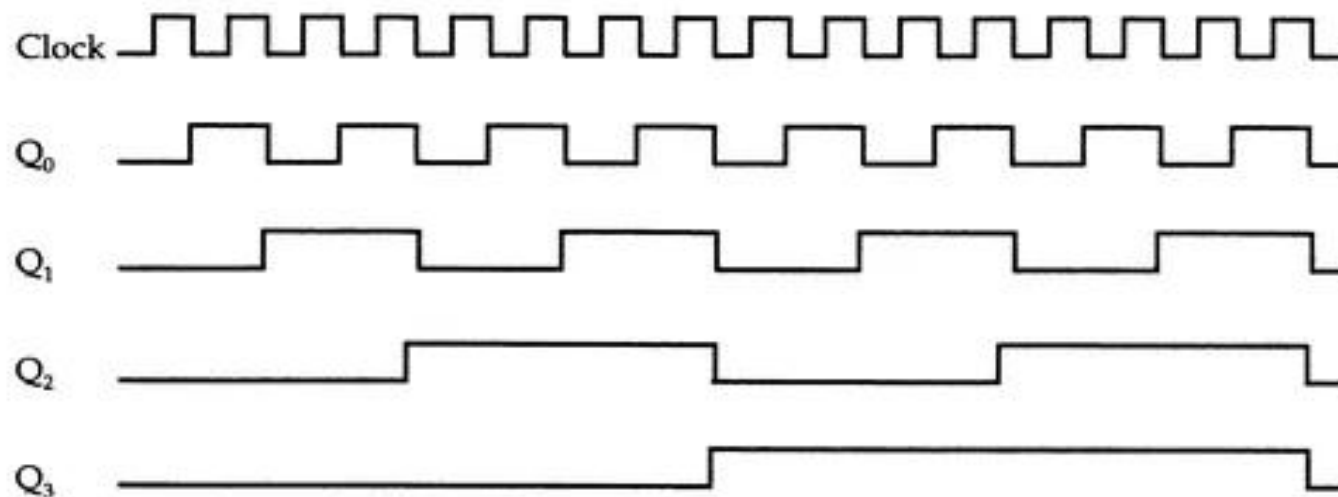


Ripple counters/Asynchronous counters

- 4 bit asynchronous up counter.



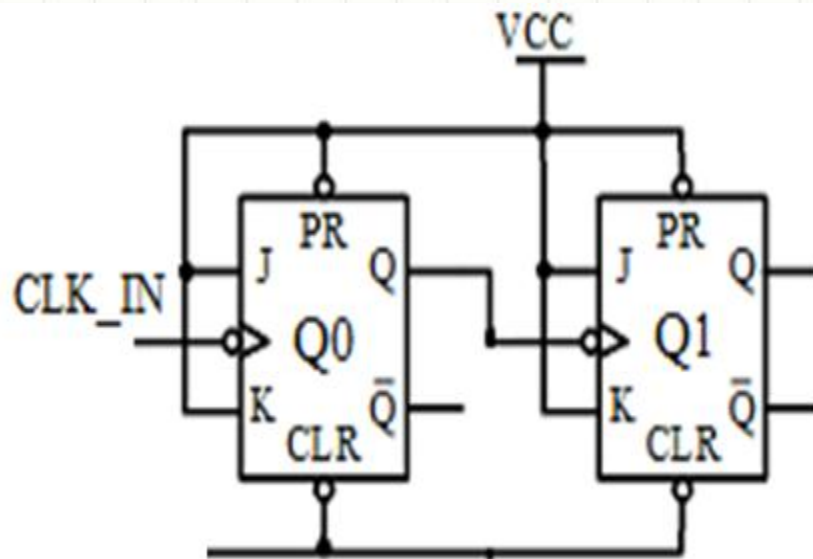
(a) Sequential Circuit



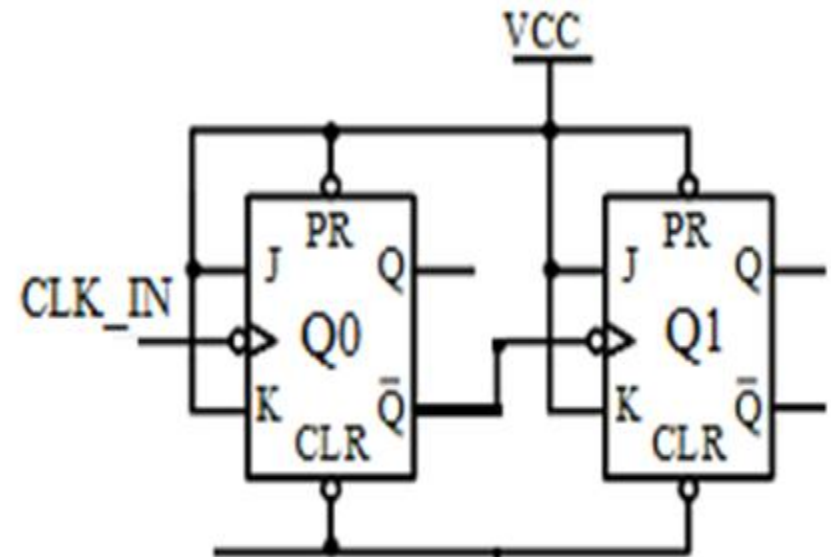
(b) Timing Diagram

Ripple counters/Asynchronous counters

- 2 bit asynchronous down counter.



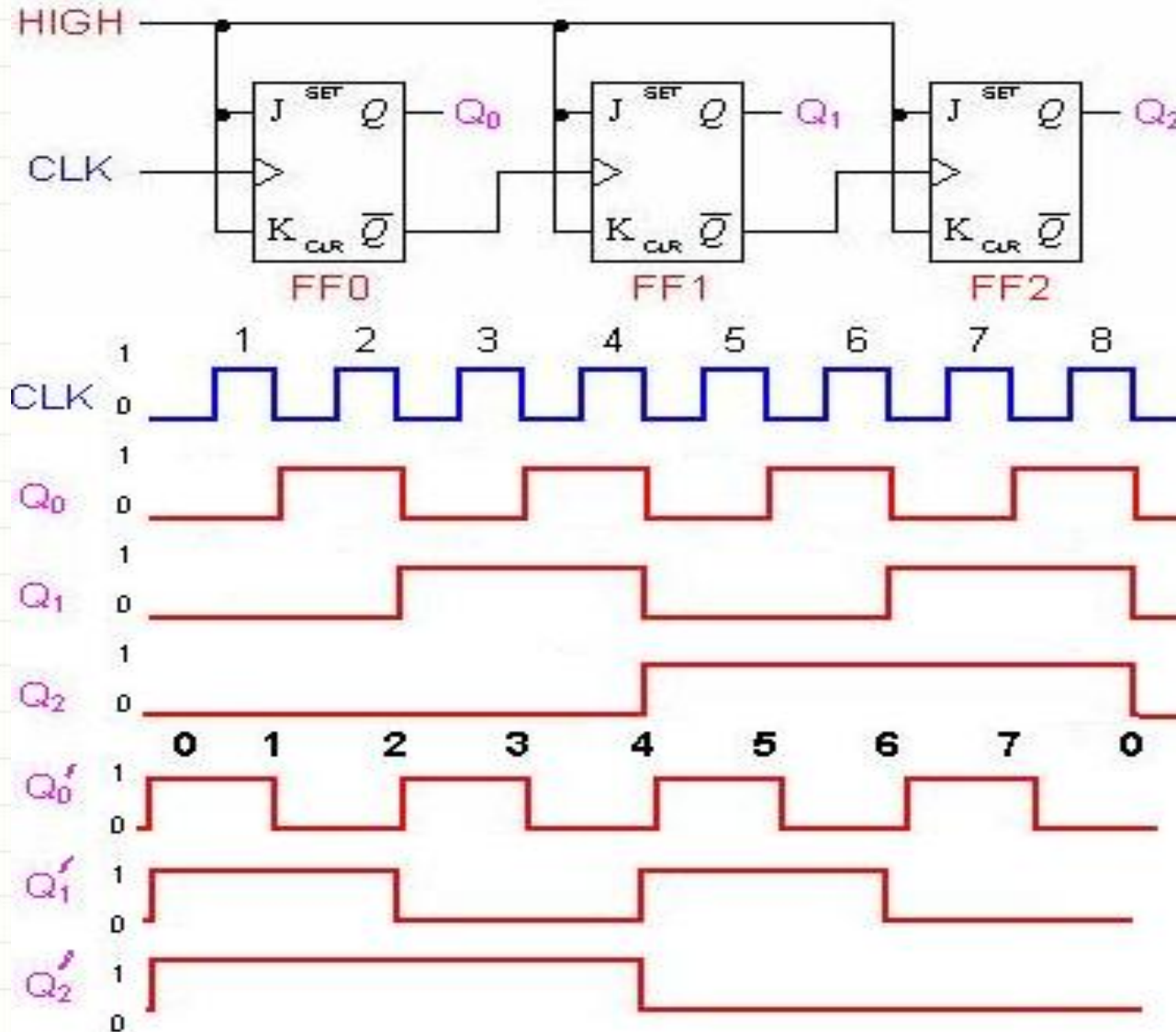
up counter



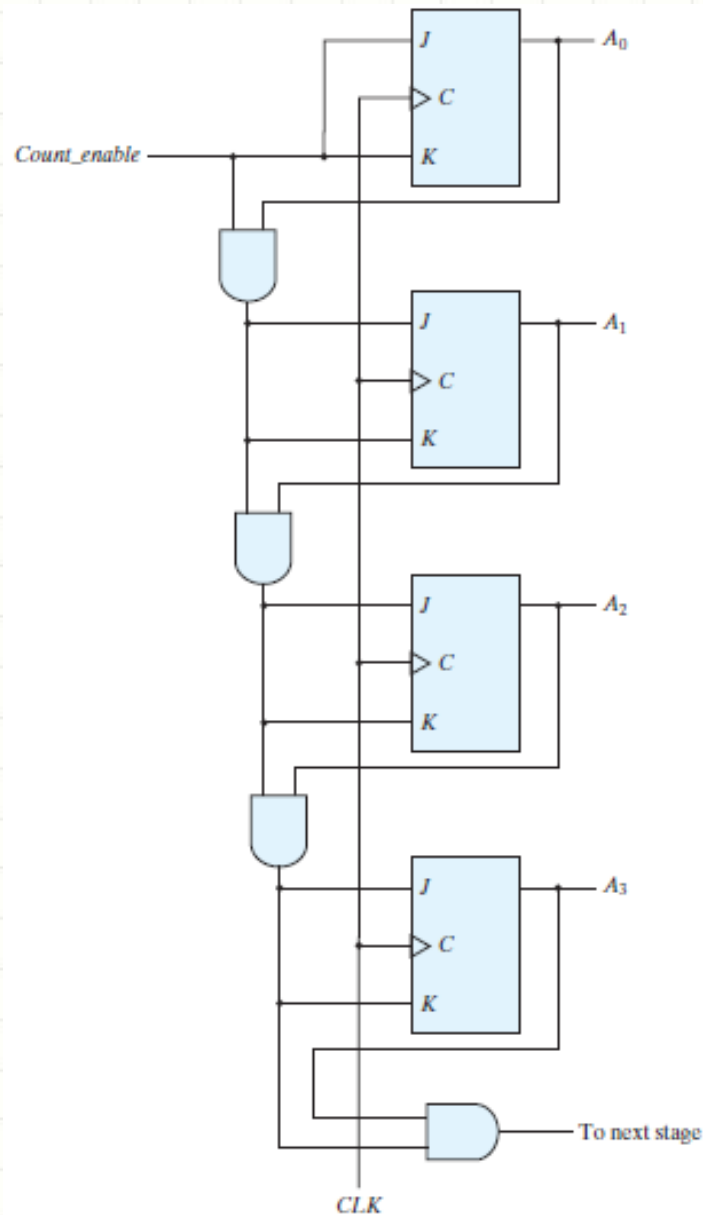
down counter

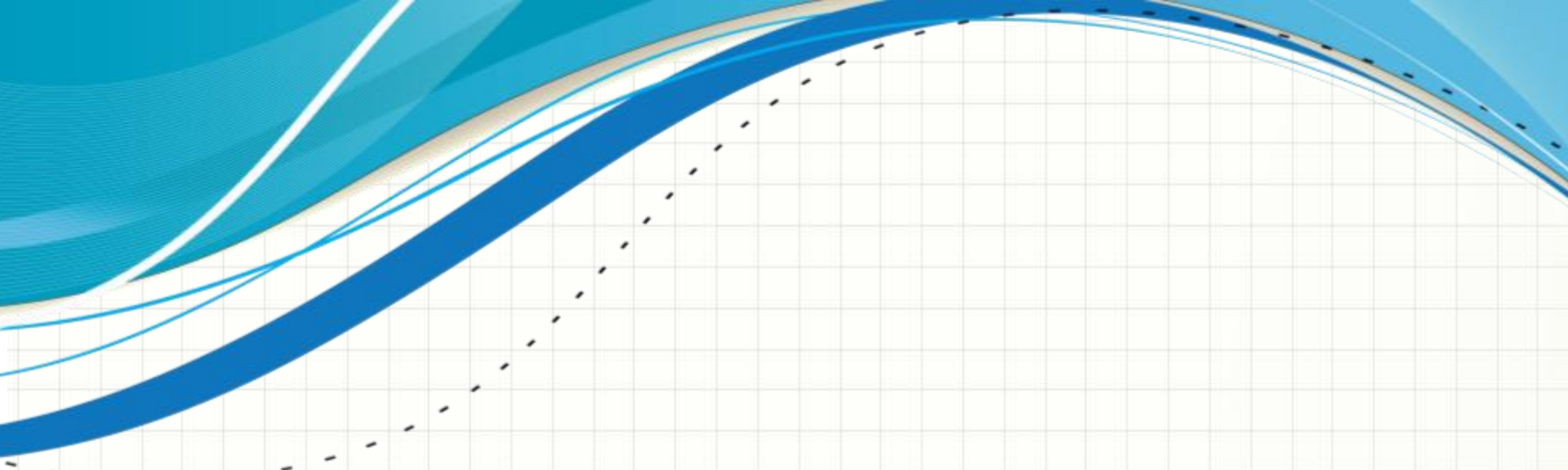
Ripple counters/Asynchronous counters

- 3 bit asynchronous down counter.



Synchronous counters





THANKS